EE323 - Analog and Digital Communications
Midterm Exam-2 Fall 2015
National University of Computer and Emerging Sciences, Lahore

Course Code: EE323

e^{-at} $e^{-a|t|} = \frac{a}{a^2+w^2}$

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Course Code: EE323	Semester: Fall 2015
September 2015	Time: 60 minutes
Student Name [redacted]	Roll number [redacted]

Q.No.1

The time autocorrelation function of a random signal is given as:

$$R(\tau) = e^{-2\alpha|\tau|} \qquad -\infty < \tau < \infty$$

a) Find the power spectral density $S(\omega)$.

b) Compute the essential bandwidth of the signal that contains 90% of the total power.

You may use following formula where needed:

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

Q. No. 2

The relationship between channel capacity C (in bits per second, bps) and bandwidth B (in Hertz, Hz) in the presence of additive white Gaussian noise (AWGN) is highlighted by Shannon's equation:

$$C = B\log_2(1 + SNR) \quad bps$$

where SNR is signal to noise ratio, the ratio between the signal power and the noise power level. Suppose that the spectrum of the communication channel is between 10 MHz and 12 MHz, and the SNR requirement for the reliable communication in the presence of noise is 24 dB.

$$Hint: SNR_{dB} = 10\log_{10} SNR$$

(a) Define bandwidth for a communication channel and then find the channel capacity.

(b) If there were no noise on the channel, how many signal levels would be needed to achieve the channel capacity found above in part (a). The Nyquist formula that relates these quantities is defined as:

$$C = 2B\log_2 M$$

where M is the signal (often voltage) levels.